MSG No. 127.389 $P4/m'bm$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m'..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,4,7,8,11,12,13,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,5,6,9,10,14,15]`

Table 4: Wyckoff site: `2d`, site symmetry: `m'.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,4,7,8,11,12,13,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,5,6,9,10,14,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[9,10,11,12]`
4	$[0, 0, -z]$	`[13,14,15,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,4,7,8]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[2,3,5,6]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[9,10,14,15]`
4	$\left[0, \frac{1}{2}, -z\right]$	`[11,12,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,8,11,16]`
2	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[2,5,10,15]`
3	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[3,6,9,14]`
4	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[4,7,12,13]`

Table 8: Wyckoff site: `4h`, site symmetry: `m'.2'm`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,8,11,16]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[2,5,10,15]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[3,6,9,14]`
4	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[4,7,12,13]`

Table 9: Wyckoff site: `8i`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,16]`
2	$[-y,\, x,\, 0]$	`[2,15]`
3	$[y,\, -x,\, 0]$	`[3,14]`
4	$[-x,\, -y,\, 0]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[5,10]`
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[6,9]`
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[7,12]`
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[8,11]`

Table 10: Wyckoff site: `8j`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,16]`
2	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[2,15]`
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[3,14]`
4	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[4,13]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[5,10]`
6	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[6,9]`
7	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[7,12]`
8	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[8,11]`

Table 11: Wyckoff site: `8k`, site symmetry: `..m`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\, x,\, z\right]$	`[2,5]`

continued ...

Table 11

No.	position	mapping
3	$\left[x+\frac{1}{2},\,-x,\,z\right]$	[3,6]
4	$\left[-x,\,\frac{1}{2}-x,\,z\right]$	[4,7]
5	$\left[x+\frac{1}{2},\,-x,\,-z\right]$	[9,14]
6	$\left[\frac{1}{2}-x,\,x,\,-z\right]$	[10,15]
7	$\left[x,\,x+\frac{1}{2},\,-z\right]$	[11,16]
8	$\left[-x,\,\frac{1}{2}-x,\,-z\right]$	[12,13]

Table 12: Wyckoff site: 16l, site symmetry: 1

No.	position	mapping
1	$[x,\,y,\,z]$	[1]
2	$[-y,\,x,\,z]$	[2]
3	$[y,\,-x,\,z]$	[3]
4	$[-x,\,-y,\,z]$	[4]
5	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,z\right]$	[5]
6	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,z\right]$	[6]
7	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,z\right]$	[7]
8	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,z\right]$	[8]
9	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,-z\right]$	[9]
10	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,-z\right]$	[10]
11	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,-z\right]$	[12]
13	$[-x,\,-y,\,-z]$	[13]
14	$[y,\,-x,\,-z]$	[14]
15	$[-y,\,x,\,-z]$	[15]
16	$[x,\,y,\,-z]$	[16]